



## Coordination Engineering for Federated Human–AI Intelligence

Open-source coordination harness. Diverse intelligence collaborates across trust boundaries at scale.  
Self-sovereign data. Formal reasoning. Cryptographic provenance. Human-in-the-loop governance.

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Org **DreamLab AI**  
Web [visionflow.info](https://visionflow.info)  
GitHub [DreamLab-AI](https://github.com/DreamLab-AI)  
Licence **VisionClaw: MPL 2.0**  
Protocol stack: **AGPL 3.0**

**5**  
Federated  
Substrates

**92**  
CUDA  
Kernels

**185+**  
MCP  
Tools

**55×**  
GPU  
Speedup

**90+**  
Agent  
Skills

**6**  
OSS  
Repos

### PROBLEM

#### Hard problems share a common shape

Climate modelling, drug discovery, creative production at scale. All require diverse intelligence collaborating across trust boundaries where centralised coordination breaks down.

**73% of frontline AI adoption happens without management sign-off.**

Uncoordinated agents waste **60–80% of tokens** on context rediscovery, duplicate reasoning, and contradictory outputs.

### WHY TOKEN VENDORS SHOULD CARE

#### Coordination is the token multiplier

- **3–5× effective throughput gain.** Shared ontology eliminates re-derivation. Provenance eliminates re-validation. Governance prevents decision spin.
- **10,000:1 problem-to-token ratio.** \$2.6B drug compounds justify \$10K–100K in coordinated reasoning if one wrong conclusion is prevented.
- **Break-even at 3 agents.** Ontology, identity spine, governance are shared infra, not per-agent costs.
- **Model-agnostic.** Any LLM behind MCP tools. Multi-vendor consumption scales with problem complexity, not lock-in.

### ARCHITECTURE

#### Five substrates, one cryptographic identity spine

Every actor (human, agent, server) shares a single `did:nostr secp256k1` keypair, verified at every layer.

**VisionClaw** | Knowledge Engineering  
OWL 2 EL reasoning, 92 CUDA kernels, 55× GPU speedup, 3D graph, 17 MCP tools

**Agentbox** | Harness Engineering  
90+ skills, 185+ MCP tools, governance relay, BIP-340 identity, Nix reproducible

**solid-pod-rs** | Cryptographic Foundation  
Rust Solid server, DID:Nostr + WAC + Web Ledgers, NIP-98 + OIDC + WebAuthn, HTTP 402

**nostr-rust-forum** | Governance UI  
Judgment Broker, Agent Control Surface (kinds 31400–31405), Leptos WASM, passkey auth

**DreamLab Edge** | Branded Deployment

React SPA + WASM forum, CF Workers edge, operator overlay, live at [dreamlab-ai.com](https://dreamlab-ai.com)

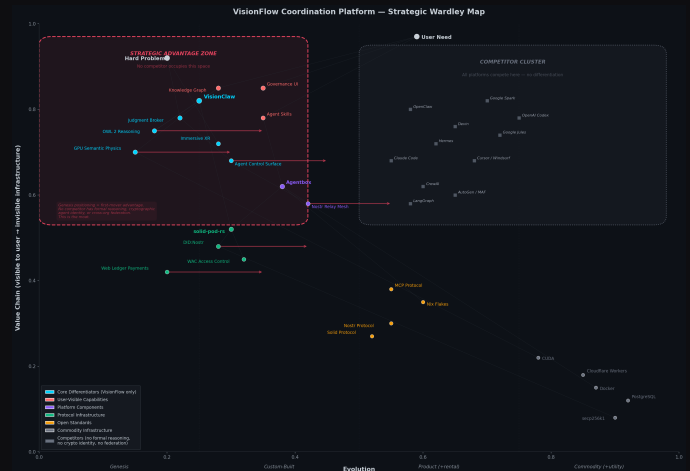
### HUMAN-IN-THE-LOOP GOVERNANCE

#### Judgment Broker

Agent publishes control panel (kind 31400) → forum renders to human → NIP-98 signed decision → VisionClaw gates knowledge mutations. Every decision is an immutable Nostr event. **90% reduction in audit reconstruction cost.**

### STRATEGIC POSITIONING: WARDLEY MAP

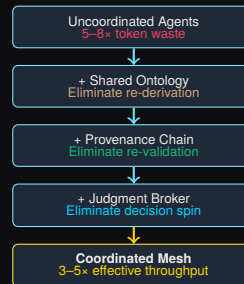
#### Every differentiator sits where no competitor can follow



VisionFlow's core differentiators (**red zone, left**) occupy the genesis/custom-built quadrant: OWL 2 formal reasoning, GPU semantic physics, Judgment Broker, cryptographic identity (DID:Nostr), immersive XR. No competitor has built here. All 11 competing platforms (grey cluster, right) sit in the commodity zone, interchangeable agent frameworks with no structural differentiation. The red dashed boundary marks the **strategic advantage zone**: capabilities that require deep protocol engineering. Cannot be replicated by wrapping an API.

### TOKEN ECONOMICS

#### Tokenmaxing for hard problems is rational



### SCALING

**Single Operator** | One API key, one container, minutes to deploy.

**Team (50+)** | Shared relay mesh, Judgment Broker oversight. Validated at scale.

**Federated Enterprise** | Sovereign nodes, trust via did:nostr. Horizontal scaling.

**Open source. Model-agnostic. Federation-ready.**

Every token produces auditable, governed, provenance-tracked reasoning.

[github.com/DreamLab-AI](https://github.com/DreamLab-AI) [visionflow.info](https://visionflow.info) [anthropic@xrsystems.uk](mailto:anthropic@xrsystems.uk)